



Organization Chart Management System

Suriaya Quddus

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ABSTRACT

Tampereen ammattikorkeakoulu
Tampere University of Applied Sciences
Bachelor Degree Programme in Software Engineering

Quddus, Suriaya

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This project refers to the development of a Java-based Organization Chart Management System using the Composite Design Pattern. The application supports building and managing a hierarchical structure of company departments and employees with add/remove functionalities and formatted visual representation. User interaction is enabled through a command-line interface, and input validation is performed using custom exception handling. The system aims to serve as an academic demonstration of object-oriented principles and user interface design.

Key words: composite design pattern, java, organization chart, command-line interface, hierarchy management

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GLOSSARY or ABBREVIATIONS AND TERMS**Abbreviation Meaning**

OOP	Object-Oriented Programming
CLI	Command-Line Interface
UML	Unified Modeling Language
IDE	Integrated Development Environment
JDK	Java Development Kit
GUI	Graphical User Interface
TAMK	Tampere University of Applied Sciences
CMD	Command Prompt (Windows Terminal)

1 INTRODUCTION

This project aimed to create a structured, maintainable Organization Chart Management System using Java. The goal was to implement features that simulate real-world department and personnel management using object-oriented principles and the Composite Design Pattern.

2 SYSTEM DESIGN

2.1 Architecture Overview

The design follows the Composite pattern to support tree-like structures. Component is the abstract base, Worker is the leaf node, and Group is the composite node holding other components. The main application logic is driven by OrganizationApp.

2.2 UML and Class Structure

- Component (abstract): common interface
- Worker: individual employee
- Group: department containing workers and sub-groups
- OrganizationApp: user interface logic
- InvalidInputException: for handling invalid name inputs

3 IMPLEMENTATION

3.1 Features Developed

- Hardcoded structure initialization with multiple departments and workers
- Dynamic addition of a worker to a department
- Dynamic removal of a worker
- Exception-based error handling and validation

3.2 UI & Functionality

- Menu-driven CLI
- Indented group/worker hierarchy with visual spacing
- Input prompts for group/worker name
- Error messages following format: "ERROR: "

4 TESTING

All menu options were tested through various scenarios including valid additions, removals, and invalid inputs. Screenshots were captured to show correct formatting, input handling, and hierarchical display. The `run.bat` script ensures smooth local execution.

5 DISCUSSION

The Composite Pattern proved to be an effective model for this use case. Key challenges included preserving formatting between parent and child nodes and enforcing valid input with minimal user friction. The CLI approach provided simplicity while validating Java skills in OOP and exception handling.

6 REFERENCES

- Gamma, E., Helm, R., Johnson, R., & Vlissides, J. (1994). Design Patterns. Addison-Wesley.
- Oracle Java Documentation. <https://docs.oracle.com/en/java/>

7 APPENDICES

Appendix 1. Source Codes, Screenshot files, and run.bat files.

GitHub Link:

<https://github.com/Suriaya-Quddus/Organization-Chart-Management-System.git>

Appendix 4. Work Hours Log Table

Date	Hours	Activity
28.04.2025	2	Initial Planning
29.04.2025	3	Implementing add or remove function
30.04.2025	2	Final UI Formatting, fixing spaces
30.04.2025	1	Create run.bat, error testing
30.04.2025	2	Final update